# Fengyifei XU

Electrical Engineer · Power Electronics & Electric Machine Drives · 800 V SiC & EMC

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M.Sc. Electrical Engineering candidate at the National University of Singapore (GPA 4.28/5.00, graduating December 2026) with a B.Eng. in Electrical Engineering and Automation from Sichuan University (GPA 3.87/4.00, top 3%). Research in wideband motor-impedance modeling for EMI/EMC, hands-on 800 V SiC partial-discharge testing as student lead of a Huawei industry collaboration, a design-institute internship on a 90 MWp solar PV + 80.2 MWh BESS bid, and an ML code-similarity Final Year Project for design-IP protection. Available full-time from January 2027.

## EDUCATION

**National University of Singapore — M.Sc. Electrical Engineering** Aug 2025 – Dec 2026 (expected)

- GPA 4.28/5.00
- Selected coursework: Electromagnetic Compatibility (A), Microwave Electronics (A), Neural Networks (A), Power Electronic Systems
- M.Sc. research: wideband induction-motor impedance modeling for EMI/EMC (MMIC Modeling and Package Laboratory, supervisor Dr. Zhenyu ZHAO)

**Sichuan University — B.Eng. Electrical Engineering and Automation** Sep 2021 – Jun 2025

*Degree conferred June 2025*

- GPA 3.87/4.00, top 3% of major; selected for the “Excellent Engineer” programme
- Coursework: High Voltage Engineering, Electric Machines, Power System Protection (Relaying), Power Electronics, Microcomputer Principles & Interfaces, Computer Networks & Communication, Operations Research (elective)

**NUS (Suzhou) Research Institute — joint undergraduate programme in Electrical & Computer Engineering** Sep 2024 – Jun 2025

*Final year of the Sichuan University B.Eng.*

- Coursework: Silicon Power Devices & Circuits, Renewable Energy Generation & Smart Grid, Machine Learning: Models & Applications
- Final Year Project (Verilog code-similarity detection) selected for the Best Projects Showcase

## RESEARCH & PROJECTS

**Wideband Induction-Motor Impedance Modeling for EMI/EMC — National University of Singapore** Aug 2025 – Aug 2026

*M.Sc. research · MMIC Modeling and Package Laboratory · supervisor Dr. Zhenyu ZHAO · dataset: github.com/Arash279/motor-impedance-db*

- Developed a physics-informed grey-box circuit model that reproduces an induction motor’s terminal impedance across **100 Hz – 100 MHz** (6 decades, 1,590 measured points) for EMI/EMC analysis and filter design in inverter-fed drives.
- Built a hybrid identification workflow (log-domain differential evolution + multi-start robust least squares) that fits the model in about **20 s** on a laptop and returns the same answer on every run (10 seeds, ±0.002%), with roughly **3× lower error** than single-algorithm variants.
- Explored the circuit structure: traced the misfit above 10 MHz to the model and added a resonant port branch that captures the high-frequency phase rise and impedance recovery.
- Helped organise the measurement data into a public dataset (github.com/Arash279/motor-impedance-db).

**Resonance-Aligned Impedance Prediction for Common-Mode Chokes — independent research** 2026 – ongoing

*Independent researcher · public NUS-EMBench data*

- Predicted the 100 kHz – 200 MHz impedance of chokes with unmeasured turn counts by aligning resonances and interpolating measured reference spectra; on **39 held-out devices** the transparent method beat the published RANN neural-network predictions (**0.083 vs 0.095** relative error, better on 27 of 39).

**High-Frequency Modeling of Inverter-Fed AC Machines: A Review — journal manuscript, under review** 2026

*Co-author with Dr. Zhenyu ZHAO*

- Wrote an engineering-oriented review (≈93 references) that organises high-frequency machine models by the output an engineer needs, the parameter source and the validation evidence, and proposes a practical model-selection framework.

**EV Motor Insulation and Partial-Discharge Testing — Sichuan University × Huawei (industry collaboration)** Sep 2023 – Dec 2023

*Student Project Lead, team of 3*

- Led a 3-person student team and contributed to building an **800 V SiC** inverter-fed motor-insulation test platform supporting online and offline testing, with automated high-frequency partial-discharge (PD) signal acquisition and data processing; performed the PD tests hands-on.
- Designed a **24-condition** experiment and built a volume–time theoretical model to analyse how pulse characteristics and overvoltage affect partial-discharge inception voltage (PDIV).
- Built a neural-network PDIV prediction model with **< 5% error**, supporting rapid insulation-performance and reliability assessment.
- Contributed to refining an industry test standard and proposed a multi-parameter PD analysis approach, supporting insulation testing for Huawei's new-energy-vehicle motor development.

**Machine-Learning Verilog Code-Similarity Detection (Final Year Project) — NUS (Suzhou) Research Institute** Sep 2024 – Jun 2025

*Independent project under supervision · selected for the Best Projects Showcase*

- Built an ML pipeline that flags similar Verilog designs for design-IP and copyright review: eight static HDL features, Gaussian-Copula + conditional-GAN augmentation of the scarce positive class (409 vs 74,282 code pairs), and a PyTorch classifier.
- Manuscript results: positive-class **F1 0.98** on a balanced test and **recall 0.97 / F1 0.86** at ≈1:48 imbalance — about **2.7× the F1 of MOSS**, the standard similarity tool.

## EXPERIENCE

**Southwest Electric Power Design Institute Co., Ltd. (China Energy Engineering Group) — Chengdu, China** Jan 2025 – Feb 2025

*Design Engineer Assistant (Intern)*

- Ran bid-stage energy-yield and system-performance simulations in PVsyst and Candela for a **90 MWp / 73.7 MWac** solar PV plant with an **80.2 MWh** battery energy storage system (BESS) in Timor-Leste.
- Selected PV modules and inverters against project capacity requirements using applicable standards and manufacturer datasheets; supported equipment sizing and capacity checks.
- Supported electrical calculations, owner technical-specification review, and CAPEX/OPEX assessment.
- Helped compile English-language material for the technical proposal.

## HONORS & AWARDS

- Provincial Gold Award, 9th Sichuan "Internet+" Innovation & Entrepreneurship Competition (2023)
- National College Student Innovation Program project rated Provincial Excellent (2023)
- College First-Class Comprehensive Scholarship, School of Electrical Engineering, Sichuan University (2023)
- Selected for the "Excellent Engineer" programme, Sichuan University; B.Eng. top 3% of major

## LEADERSHIP & VOLUNTEERING

**31st FISU Summer World University Games, Chengdu** 2023

*Volunteer & Student Work-Group Leader*

- Coordinated volunteer tasks and resource scheduling, liaised on foreign-affairs matters and with international student volunteers, prepared contingency plans and handled incidents on site.

## SKILLS

**Electrical engineering:** electric machines & motor drives, power electronics, EMI/EMC analysis, common-/differential-mode (CM/DM) impedance analysis, equivalent-circuit modeling, frequency-response analysis, high-voltage insulation & partial-discharge testing, PV + BESS system design basics

**Simulation:** MATLAB/Simulink (incl. feedback-control tuning), Multisim, PVsyst, Candela

**Lab & instruments:** oscilloscope, impedance analyzer, high-voltage / partial-discharge test setups

**Data & programming:** Python (data processing, numerical optimisation, neural-network models), C/C++, SQLite, Excel (advanced formulas, macros, pivot tables), machine learning (class-imbalance handling, ROC-based thresholding)

**Certifications & tests:** IELTS 7.0; CET-6 578; CET-4 604

**Languages:** Mandarin (native), English (IELTS 7.0), Japanese (conversational), German (beginner)